4(a)	• particles are in (continuous) random motion • particles have negligible volume (compared with the gas) • negligible forces between particles (except during collisions) • (all) collisions (perfectly) elastic • time of collision negligible (in comparison with time between collisions) *Any two points, 1 mark each*	**B2**
4(b)(i)	(general starting equation) $pV = nRT$	**C1**
	$T = (2pV / nR)$ where R is the (molar) gas constant	**A1**
4(b)(ii)	sketch: straight vertical line XY from (V, 2p) to (V, p)	**B1**
	straight horizontal line YZ from (V, p) to (2V, p)	**B1**
	curve with gradient increasing from Z to X from (2V, p) to (V, 2p)	**B1**
4(b)(iii)	XY work done on gas correct (= 0)	**B1**
	ZX increase in internal energy correct (= 0)	**B1**
	YZ work done on gas correct (= $-pV$)	**B1**
	XY increase in internal energy such that the increase in internal energy column adds up to zero	**B1**
	all three thermal energies transferred such that $\Delta U = q + w$ in each row (completely correct answer: <table><tr><th>change</th><th>ΔU</th><th>Q</th><th>w</th></tr><tr><td>X to Y</td><td>$-U$</td><td>$-U$</td><td>0</td></tr><tr><td>Y to Z</td><td>[$+U$]</td><td>$U + pV$</td><td>$-pV$</td></tr><tr><td>Z to X</td><td>0</td><td>$-W$</td><td>[$+W$]</td></tr></table>)	**B1**